

# Circular Economy Ontology Network (CEON) - Actor ODP

## Metadata

**IRI**

<http://w3id.org/CEON/ontology/actorODP/>

**Title**

Circular Economy Ontology Network (CEON) - Actor ODP

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**Date Created**

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**Version Iri**

<http://w3id.org/CEON/ontology/actorODP/0.2/>

**Version Info**

0.2

**Preferred Namespace Prefix**

actorODP

**Preferred Namespace Uri**

<http://w3id.org/CEON/ontoloy/actorODP/>

**Description**

A core ODP of the CEON ontology network, defining aspects of the actor concept.

**Covers Requirements**

Covers the following requirements from Onto-DESIDE D3.1: CVN-Process-3, CVN-Actor-1,4,6,7, CVN-Competency-3, CVN-Information-4, C11-1, C11-3, E1-6,6,6,9, E4-10

## Classes

|                    |                                                                                                         |
|--------------------|---------------------------------------------------------------------------------------------------------|
| Actor <sup>c</sup> |                                                                                                         |
| IRI                | <a href="http://w3id.org/CEON/ontology/actorODP/Actor">http://w3id.org/CEON/ontology/actorODP/Actor</a> |
| In Domain Of       | <a href="#">actorType</a> <sup>op</sup><br><a href="#">hasActorLocation</a> <sup>op</sup>               |
| In Range Of        | <a href="#">capabilityOf</a> <sup>op</sup><br><a href="#">participatingActor</a> <sup>op</sup>          |

|                         |                                                                                                                 |
|-------------------------|-----------------------------------------------------------------------------------------------------------------|
| Actor Type <sup>c</sup> |                                                                                                                 |
| IRI                     | <a href="http://w3id.org/CEON/ontology/actorODP/ActorType">http://w3id.org/CEON/ontology/actorODP/ActorType</a> |
| In Range Of             | <a href="#">actorType</a> <sup>op</sup>                                                                         |

## Capability<sup>c</sup>

|                            |                                                                                                                   |
|----------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/actor0DP/Capability">http://w3id.org/CE0N/ontology/actor0DP/Capability</a> |
| <b><u>In Domain Of</u></b> | <a href="#">capabilityOf<sup>op</sup></a><br><a href="#">capabilityProperty<sup>op</sup></a>                      |
| <b><u>In Range Of</u></b>  | <a href="#">basedOnCapability<sup>op</sup></a>                                                                    |

## Collaboration Participation<sup>c</sup>

|                            |                                                                                                                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/actor0DP/CollaborationParticipation">http://w3id.org/CE0N/ontology/actor0DP/CollaborationParticipation</a> |
| <b><u>Sub Class Of</u></b> | <a href="#">Participation<sup>c</sup></a>                                                                                                         |
| <b><u>In Domain Of</u></b> | <a href="#">basedOnCapability<sup>op</sup></a><br><a href="#">participationIn<sup>op</sup></a>                                                    |

## Participation<sup>c</sup>

|                              |                                                                                                                                                                                                                                                                    |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                   | <a href="http://w3id.org/CE0N/ontology/actor0DP/Participation">http://w3id.org/CE0N/ontology/actor0DP/Participation</a>                                                                                                                                            |
| <b><u>Sub Class Of</u></b>   | <a href="#">participationStartTime<sup>dp</sup></a> <i>exactly 1</i> <a href="#">1<sup>c</sup></a> <i>or</i> <a href="#">participationTimePoint<sup>dp</sup></a> <i>exactly 1</i> <a href="#">1<sup>c</sup></a>                                                    |
| <b><u>In Domain Of</u></b>   | <a href="#">participantRole<sup>op</sup></a><br><a href="#">participatingActor<sup>op</sup></a><br><a href="#">participationEndTime<sup>dp</sup></a><br><a href="#">participationStartTime<sup>dp</sup></a><br><a href="#">participationTimePoint<sup>dp</sup></a> |
| <b><u>Super Class Of</u></b> | <a href="#">CollaborationParticipation<sup>c</sup></a><br><a href="#">ResourceParticipation<sup>c</sup></a><br><a href="#">ResourceRelation<sup>c</sup></a>                                                                                                        |

## Resource Participation<sup>c</sup>

|                            |                                                                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/actor0DP/ResourceParticipation">http://w3id.org/CE0N/ontology/actor0DP/ResourceParticipation</a> |
| <b><u>Sub Class Of</u></b> | <a href="#">Participation<sup>c</sup></a>                                                                                               |
| <b><u>In Domain Of</u></b> | <a href="#">participatingObject<sup>op</sup></a><br><a href="#">participatingSubject<sup>op</sup></a>                                   |

## Resource Relation<sup>c</sup>

|                            |                                                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/actor0DP/ResourceRelation">http://w3id.org/CE0N/ontology/actor0DP/ResourceRelation</a> |
| <b><u>Sub Class Of</u></b> | <a href="#">Participation<sup>c</sup></a>                                                                                     |
| <b><u>In Domain Of</u></b> | <a href="#">participatingResource<sup>op</sup></a>                                                                            |

## Role<sup>c</sup>

**IRI** <http://w3id.org/CE0N/ontology/actor0DP/Role>

**In Range Of** [participantRole](#)<sup>op</sup>

## Collaboration<sup>c</sup>

**IRI** <http://w3id.org/CE0N/ontology/cvn/Collaboration>

## Process<sup>c</sup>

**IRI** <http://w3id.org/CE0N/ontology/process0DP/Process>

## Resource<sup>c</sup>

**IRI** <http://w3id.org/CE0N/ontology/resource0DP/Resource>

**In Range Of** [participatingResource](#)<sup>op</sup>

## Geometry<sup>c</sup>

**IRI** <http://www.opengis.net/ont/geosparql#Geometry>

**Description** A coherent set of direct positions in space. The positions are held within a Spatial Reference System (SRS).

**In Domain Of**

- [as GML](#)<sup>dp</sup>
- [as GeoJSON](#)<sup>dp</sup>
- [as WKT](#)<sup>dp</sup>

**In Range Of** [hasActorLocation](#)<sup>op</sup>

## Object Properties

### actor type<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/actor0DP/actorType>

**Domain** [Actor](#)<sup>c</sup>

**Range** [ActorType](#)<sup>c</sup>

## based on capability<sup>op</sup>

|               |                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/actor0DP/basedOnCapability">http://w3id.org/CE0N/ontology/actor0DP/basedOnCapability</a> |
| <b>Domain</b> | <a href="#">CollaborationParticipation</a> <sup>c</sup>                                                                         |
| <b>Range</b>  | <a href="#">Capability</a> <sup>c</sup>                                                                                         |

## capability of<sup>op</sup>

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/actor0DP/capabilityOf">http://w3id.org/CE0N/ontology/actor0DP/capabilityOf</a> |
| <b>Domain</b> | <a href="#">Capability</a> <sup>c</sup>                                                                               |
| <b>Range</b>  | <a href="#">Actor</a> <sup>c</sup>                                                                                    |

## capability property<sup>op</sup>

|               |                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/actor0DP/capabilityProperty">http://w3id.org/CE0N/ontology/actor0DP/capabilityProperty</a> |
| <b>Domain</b> | <a href="#">Capability</a> <sup>c</sup>                                                                                           |

## has actor location<sup>op</sup>

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/actor0DP/hasActorLocation">http://w3id.org/CE0N/ontology/actor0DP/hasActorLocation</a> |
| <b>Domain</b> | <a href="#">Actor</a> <sup>c</sup>                                                                                            |
| <b>Range</b>  | <a href="#">Geometry</a> <sup>c</sup>                                                                                         |

## participant role<sup>op</sup>

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/actor0DP/participantRole">http://w3id.org/CE0N/ontology/actor0DP/participantRole</a> |
| <b>Domain</b> | <a href="#">Participation</a> <sup>c</sup>                                                                                  |
| <b>Range</b>  | <a href="#">Role</a> <sup>c</sup>                                                                                           |

## participating actor<sup>op</sup>

|               |                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/actor0DP/participatingActor">http://w3id.org/CE0N/ontology/actor0DP/participatingActor</a> |
| <b>Domain</b> | <a href="#">Participation</a> <sup>c</sup>                                                                                        |
| <b>Range</b>  | <a href="#">Actor</a> <sup>c</sup>                                                                                                |

## participating object<sup>op</sup>

|               |                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/actor0DP/participatingObject">http://w3id.org/CE0N/ontology/actor0DP/participatingObject</a> |
| <b>Domain</b> | <a href="#">ResourceParticipation</a> <sup>c</sup>                                                                                  |

## participating resource<sup>op</sup>

|               |                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/actor0DP/participatingResource">http://w3id.org/CE0N/ontology/actor0DP/participatingResource</a> |
| <b>Domain</b> | <a href="#">ResourceRelation<sup>c</sup></a>                                                                                            |
| <b>Range</b>  | <a href="#">resourceODP:Resource<sup>c</sup></a>                                                                                        |

## participating subject<sup>op</sup>

|               |                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/actor0DP/participatingSubject">http://w3id.org/CE0N/ontology/actor0DP/participatingSubject</a> |
| <b>Domain</b> | <a href="#">ResourceParticipation<sup>c</sup></a>                                                                                     |

## participation in<sup>op</sup>

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/actor0DP/participationIn">http://w3id.org/CE0N/ontology/actor0DP/participationIn</a> |
| <b>Domain</b> | <a href="#">CollaborationParticipation<sup>c</sup></a>                                                                      |
| <b>Range</b>  | <a href="#">processODP:Process<sup>c</sup></a> or <a href="#">cvn:Collaboration<sup>c</sup></a>                             |

## Datatype Properties

### participation end time<sup>dp</sup>

|               |                                                                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/actor0DP/participationEndTime">http://w3id.org/CE0N/ontology/actor0DP/participationEndTime</a>                                   |
| <b>Domain</b> | <a href="#">Participation<sup>c</sup></a>                                                                                                                               |
| <b>Range</b>  | <a href="#">xsd:dateTime<sup>c</sup></a> or <a href="#">xsd:date<sup>c</sup></a> or <a href="#">xsd:gMonthYear<sup>c</sup></a> or <a href="#">xsd:gYear<sup>c</sup></a> |

### participation start time<sup>dp</sup>

|               |                                                                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/actor0DP/participationStartTime">http://w3id.org/CE0N/ontology/actor0DP/participationStartTime</a>                               |
| <b>Domain</b> | <a href="#">Participation<sup>c</sup></a>                                                                                                                               |
| <b>Range</b>  | <a href="#">xsd:date<sup>c</sup></a> or <a href="#">xsd:gYear<sup>c</sup></a> or <a href="#">xsd:gMonthYear<sup>c</sup></a> or <a href="#">xsd:dateTime<sup>c</sup></a> |

### participation time point<sup>dp</sup>

|               |                                                                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/actor0DP/participationTimePoint">http://w3id.org/CE0N/ontology/actor0DP/participationTimePoint</a>                               |
| <b>Domain</b> | <a href="#">Participation<sup>c</sup></a>                                                                                                                               |
| <b>Range</b>  | <a href="#">xsd:gYear<sup>c</sup></a> or <a href="#">xsd:gMonthYear<sup>c</sup></a> or <a href="#">xsd:date<sup>c</sup></a> or <a href="#">xsd:dateTime<sup>c</sup></a> |

as GML<sup>dp</sup>

|                      |                                                                                                                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>           | <a href="http://www.opengis.net/ont/geosparql#asGML">http://www.opengis.net/ont/geosparql#asGML</a>                                                                                             |
| <b>Is Defined By</b> | <a href="http://www.opengis.net/spec/geosparql/1.0/req/geometry-extension/geometry-as-wkt-literal">http://www.opengis.net/spec/geosparql/1.0/req/geometry-extension/geometry-as-wkt-literal</a> |
| <b>Description</b>   | The GML serialization of a Geometry.                                                                                                                                                            |
| <b>Domain</b>        | <a href="#">Geometry</a> <sup>c</sup>                                                                                                                                                           |
| <b>Range</b>         | <a href="#">geo:gmlLiteral</a>                                                                                                                                                                  |

as GeoJSON<sup>dp</sup>

|                      |                                                                                                                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>           | <a href="http://www.opengis.net/ont/geosparql#asGeoJSON">http://www.opengis.net/ont/geosparql#asGeoJSON</a>                                                                                     |
| <b>Is Defined By</b> | <a href="http://www.opengis.net/spec/geosparql/1.0/req/geometry-extension/geometry-as-wkt-literal">http://www.opengis.net/spec/geosparql/1.0/req/geometry-extension/geometry-as-wkt-literal</a> |
| <b>Description</b>   | The GeoJSON serialization of a Geometry.                                                                                                                                                        |
| <b>Domain</b>        | <a href="#">Geometry</a> <sup>c</sup>                                                                                                                                                           |
| <b>Range</b>         | <a href="#">geo:geoJSONLiteral</a>                                                                                                                                                              |

as WKT<sup>dp</sup>

|                      |                                                                                                                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>           | <a href="http://www.opengis.net/ont/geosparql#asWKT">http://www.opengis.net/ont/geosparql#asWKT</a>                                                                                             |
| <b>Is Defined By</b> | <a href="http://www.opengis.net/spec/geosparql/1.0/req/geometry-extension/geometry-as-wkt-literal">http://www.opengis.net/spec/geosparql/1.0/req/geometry-extension/geometry-as-wkt-literal</a> |
| <b>Description</b>   | The WKT serialization of a Geometry.                                                                                                                                                            |
| <b>Domain</b>        | <a href="#">Geometry</a> <sup>c</sup>                                                                                                                                                           |
| <b>Range</b>         | <a href="#">geo:wktLiteral</a>                                                                                                                                                                  |

## Annotation Properties

contributor<sup>ap</sup>

|            |                                                                                         |
|------------|-----------------------------------------------------------------------------------------|
| <b>IRI</b> | <a href="http://purl.org/dc/terms/contributor">http://purl.org/dc/terms/contributor</a> |
|------------|-----------------------------------------------------------------------------------------|

created<sup>ap</sup>

|            |                                                                                 |
|------------|---------------------------------------------------------------------------------|
| <b>IRI</b> | <a href="http://purl.org/dc/terms/created">http://purl.org/dc/terms/created</a> |
|------------|---------------------------------------------------------------------------------|

creator<sup>ap</sup>

|            |                                                                                 |
|------------|---------------------------------------------------------------------------------|
| <b>IRI</b> | <a href="http://purl.org/dc/terms/creator">http://purl.org/dc/terms/creator</a> |
|------------|---------------------------------------------------------------------------------|

|                                          |                                                                                                                                                                                                 |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| description <sup>ap</sup>                |                                                                                                                                                                                                 |
| IRI                                      | <a href="http://purl.org/dc/terms/description">http://purl.org/dc/terms/description</a>                                                                                                         |
| license <sup>ap</sup>                    |                                                                                                                                                                                                 |
| IRI                                      | <a href="http://purl.org/dc/terms/license">http://purl.org/dc/terms/license</a>                                                                                                                 |
| title <sup>ap</sup>                      |                                                                                                                                                                                                 |
| IRI                                      | <a href="http://purl.org/dc/terms/title">http://purl.org/dc/terms/title</a>                                                                                                                     |
| preferred namespace prefix <sup>ap</sup> |                                                                                                                                                                                                 |
| IRI                                      | <a href="http://purl.org/vocab/vann/preferredNamespacePrefix">http://purl.org/vocab/vann/preferredNamespacePrefix</a>                                                                           |
| preferred namespace uri <sup>ap</sup>    |                                                                                                                                                                                                 |
| IRI                                      | <a href="http://purl.org/vocab/vann/preferredNamespaceUri">http://purl.org/vocab/vann/preferredNamespaceUri</a>                                                                                 |
| covers requirements <sup>ap</sup>        |                                                                                                                                                                                                 |
| IRI                                      | <a href="http://www.ontologydesignpatterns.org/schemas/cpannotation-schema.owl#coversRequirements">http://www.ontologydesignpatterns.org/schemas/cpannotation-schema.owl#coversRequirements</a> |
| definition <sup>ap</sup>                 |                                                                                                                                                                                                 |
| IRI                                      | <a href="http://www.w3.org/2004/02/skos/core#definition">http://www.w3.org/2004/02/skos/core#definition</a>                                                                                     |
| pref label <sup>ap</sup>                 |                                                                                                                                                                                                 |
| IRI                                      | <a href="http://www.w3.org/2004/02/skos/core#prefLabel">http://www.w3.org/2004/02/skos/core#prefLabel</a>                                                                                       |

## Namespaces

: <http://w3id.org/CEON/ontology/actorODP/>  
 cvn <http://w3id.org/CEON/ontology/cvn/>  
 dcterms <http://purl.org/dc/terms/>  
 geo <http://www.opengis.net/ont/geosparql#>  
 odp <http://www.ontologydesignpatterns.org/schemas/cpannotationschema.owl#>  
 owl <http://www.w3.org/2002/07/owl#>

**processODP**

<http://w3id.org/CEON/ontology/processODP/>

**prov**

<http://www.w3.org/ns/prov#>

**rdf**

<http://www.w3.org/1999/02/22-rdf-syntax-ns#>

**rdfs**

<http://www.w3.org/2000/01/rdf-schema#>

**resourceODP**

<http://w3id.org/CEON/ontology/resourceODP/>

**skos**

<http://www.w3.org/2004/02/skos/core#>

**vann**

<http://purl.org/vocab/vann/>

**xsd**

<http://www.w3.org/2001/XMLSchema#>

## Legend

|    |                          |
|----|--------------------------|
| c  | Classes                  |
| op | Object<br>Properties     |
| dp | Datatype<br>Properties   |
| ap | Annotation<br>Properties |